A Comparative Analysis of XAI Techniques for Medical Imaging: Challenges and Opportunities

Paola Barra
Department of Science and Technology
University of Naples Parthenope
Naples, Italy
0000-0002-7692-0626

Attilio Della Greca
Department of Computer Science
University of Salerno
Salerno, Italy
0000-0002-4900-8666

Ilaria Amaro
Department of Computer Science
University of Salerno
Salerno, Italy
0009-0003-0592-2389

Augusto Tortora
Department of Physics "E. Pancini"
University of Naples Federico II
Naples, Italy
0000-0002-0596-906X

Mariacarla Staffa
Department of Science and Technology
University of Naples Parthenope
Naples, Italy
0000-0001-7656-8370

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***Abstract*—The application of artificial intelligence (AI) in medical imaging has significantly improved diagnostic accuracy. However, the reliance on black-box models remains a barrier to its widespread adoption in clinical settings. This article compares the main techniques of explainable AI (XAI), such as Grad-CAM, LIME, and SHAP, evaluating their effectiveness in interpreting deep learning models used in medical imaging. Two case studies are analyzed, comparing the three methods and highlighting their strengths and weaknesses. The results of this analysis show that Grad-CAM provides intuitive visualizations; LIME offers excellent flexibility in application; and SHAP delivers complete and accurate explanations, despite its high computational load.**

***Index Terms*—Explainable AI, Medical imaging, Grad-CAM, LIME, SHAP**

I. Introduction

Artificial Intelligence (AI) has revolutionized medical imaging in recent years, with deep learning models often surpassing human performance in specific diagnostic tasks. Despite their remarkable accuracy, the black-box nature of these models poses a critical challenge for clinical adoption. Transparency and interpretability are essential for clinicians to trust and reliably incorporate AI-driven decisions into diagnostic workflows. Explainable AI (XAI) seeks to address this issue by providing interpretable insights into the decision-making processes of these models.

This study aims to explore and evaluate the practical applicability of three widely used XAI techniques— Grad-CAM, LIME, and SHAP— in medical imaging. By focusing on two clinically significant use cases, breast cancer and lung tumor detection, this research seeks to:

- Compare the strengths and limitations of these XAI methods in real-world diagnostic scenarios.
- Identify how each technique aligns with the interpretability and usability needs of clinicians.
- Offer evidence-based insights to guide the integration of XAI tools into routine medical imaging workflows.

By addressing these objectives, this paper contributes to the growing body of knowledge on XAI and its potential to enhance transparency, trust, and efficacy in AI-assisted diagnostics.

The structure of the paper is as follows: In the section II, we provide an overview of the most widely used XAI techniques in medical imaging. The section III offers a detailed comparison of the three leading techniques: Grad-CAM, LIME, and SHAP. In the section IV, we explore applying these methods across two specific use cases. The section V presents an analysis of the results obtained. The section VI outlines potential research opportunities in XAI for medical imaging. Finally, the section VII summarizes the essential findings and conclusions.

II. Explainable AI in Medical Imaging

Explainable AI (XAI) refers to techniques aimed at improving the interpretability and transparency of AI models, helping users understand, trust, and effectively use the models' predictions. This is especially relevant in sensitive domains like medical imaging, where understanding the AI's reasoning is crucial for clinical decision-making. XAI systems aim to balance the following two aspects: achieving high predictive accuracy and providing insights into the logic behind the AI's decisions [6], [7].

XAI methods are developed using two main categories: transparent models and post-hoc interpretability techniques. Transparent models, such as decision trees or K-Nearest Neighbors, are inherently easy to interpret because their operations align closely with human logic [4]. However, more complex models—like deep learning architectures (e.g., CNNs and RNNs)—require post-hoc methods to make their predictions understandable [5], [8].

Post-hoc interpretability is divided further into global and local explanations. Global explanations provide a comprehensive view of the entire model's behavior, covering aspects

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like its architecture and data usage. Local explanations, on the other hand, focus on the reasoning behind individual predictions by analyzing the influence of specific features.

There are two key types of post-hoc techniques:

- **Model-specific methods**: Designed for specific architectures, such as Grad-CAM for convolutional networks.
- **Model-agnostic methods**: More general and applicable to various models, like LIME and SHAP, often used to explain predictions from black-box models.

XAI is essential for improving trust in AI tools, identifying causal patterns in data, and ensuring ethical transparency, especially in applications like healthcare. This added transparency empowers medical professionals to incorporate AI solutions confidently, balancing human insight with computational power.

Several XAI techniques have emerged to enhance the interpretability of AI models in medical imaging, each contributing valuable insights to the field:

- Gradient-weighted Class Activation Mapping (Grad-CAM) [12] is a popular method that generates visual heatmaps to highlight the regions of an image that most influence a model's predictions, thereby providing intuitive insights into its decision-making process.
- Local Interpretable Model-agnostic Explanations (LIME) [14] offers a complementary approach by perturbing the input data to create local models around specific predictions, allowing clinicians to understand the contribution of individual features.
- SHapley Additive exPlanations (SHAP) [13] applies concepts from game theory to quantify the impact of each feature on a model's output, yielding precise and mathematically sound explanations.
- Integrated Gradients [15] computes the integral of gradients along a path from a baseline input to the actual input, effectively attributing the model's predictions to individual features while ensuring consistency in the explanations.
- Activation Maximization [16] generates images that maximize the activation of specific neurons in a neural network, allowing researchers to visualize and understand the learned features that contribute to the model's predictions.
- Attention Mechanisms [10], often integrated into neural networks, highlight the parts of an input image that the model focuses on during prediction, providing insights into its reasoning process.
- Saliency Maps [11] visualize the gradient of the output with respect to the input image, identifying which pixels significantly impact the model's decisions, thereby illustrating areas of importance in medical images.
- Counterfactual Explanations [2] generates scenarios where input features are altered to show how these changes would affect the model's predictions, helping to clarify decision boundaries and reasoning.
- Example-based Explanations [9] help users understand the model's decision in the context of analogous cases, by providing instances from the training data that are similar to a specific prediction.
- Feature Visualization [3] aims to visualize the features that the model has learned, providing insights into the internal representations and what aspects of the input are deemed significant for predictions.

In this article we will concentrate our analysis on Grad-CAM, LIME, and SHAP due to their widespread adoption and recognition in both research and clinical settings [17]–[19]. These three techniques not only represent a diverse range of explanation methodologies—visual, local, and quantitative—but also collectively address many of the challenges faced in interpreting AI models. By providing a comparative analysis of these approaches, we aim to highlight their strengths and weaknesses, ultimately enhancing our understanding of improving AI interpretability in medical imaging.

III. Comparison of Techniques

In this section, the characteristics of the three considered XAI techniques are described.

A. Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) [12] is a widely used technique to visually interpret decisions made by convolutional neural networks (CNNs), particularly in tasks like medical image analysis. It highlights regions in an input image that most influence the model's output, making it useful for applications such as detecting abnormalities in radiographs, Magnetic resonance imaging (MRIs), or mammograms.

The core idea behind Grad-CAM is to use the gradients of a specific class for the activations in the final convolutional layer. These gradients reveal which parts of the image contribute the most to the model's decision. By weighting the activation maps with the average of these gradients, Grad-CAM creates a heatmap showing relevant regions. The ReLU function visualises only optimistic influences, discarding any negative activations to improve interpretability.

Grad-CAM offers several advantages: it works across different CNN architectures without requiring structural changes and provides class-discriminative insights. However, its heatmaps often need to be higher resolution, limiting the precision needed for clinical applications.

This method bridges the technical complexity of deep learning models and practical interpretability, offering clinicians a visual tool to better understand predictions. However, Grad-CAM also has limitations—while it offers visual explanations, these may not always provide the detailed granularity necessary for certain medical decisions.

B. LIME

Local Interpretable Model-Agnostic Explanations (LIME) [14] is a model-agnostic interpretability technique that generates explanations for individual predictions by analyzing the local behavior of a machine learning model. The core idea

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is that while global model behavior might be complex, the local behavior near a single input can be approximated with interpretable models like linear regression.

LIME operates by perturbing the input data (e.g., altering image regions or text snippets) to observe how these small changes influence the model's predictions. For image classification tasks, LIME can work with interpretable representations such as "super-pixels", highlighting sections of the image most relevant to the model's decision. It fits a simple linear model to the perturbed data around the input instance, helping to explain why the original model arrived at a particular outcome.

A key feature of LIME is that it works across various models—whether neural networks or tree-based models—without requiring structural changes, making it highly flexible. However, the consistency of LIME's explanations can vary, as they depend on the random sampling process used for perturbations. The method also introduces a trade-off between interpretability and fidelity; while simpler models are easier to understand, they may not fully capture the complexity of the original prediction.

LIME assigns higher weights to samples closer to the instance being explained using a kernel function (often a radial basis function), ensuring that the local model remains faithful to the original prediction. This sparse, linear explanation method emphasizes a limited set of features to maintain interpretability, often employing regularization techniques like Lasso to select relevant features. Overall, LIME provides intuitive insights into black-box models but demands careful tuning to ensure reliability across different applications.

C. SHAP

Shapley Additive exPlanations (SHAP) [13] is an interpretability technique grounded in game theory, designed to assign importance scores—called Shapley values—to individual features of a machine learning model's prediction. Each feature's SHAP value reflects its contribution to the final prediction, offering a mathematically principled way to explain both local (instance-level) and global (model-level) behavior. The method ensures that the sum of the SHAP values equals the model's output for the analysed instance.

The approach relies on evaluating how much each feature, alone and in combination with others, influences the model's output. This process resembles dividing a payout fairly among multiple players based on their contributions in a cooperative game. In the case of models, each feature is treated like a "player" whose marginal contribution is assessed across all possible feature combinations. For example, in medical imaging, SHAP assigns contributions to each pixel, making it useful for tasks like analyzing Computed Tomography (CT) scans or ultrasounds.

How SHAP works:

- Local Accuracy: Ensures that the sum of SHAP values matches the model's output for a given instance.
- Missingness Principle: Features absent in an instance are assigned a zero contribution.
- Consistency Principle: If the inclusion of a feature improves the model's output, its SHAP value should not decrease.

IV. In-depth evaluation of Grad-CAM, SHAP, and LIME across two studies

In this section, we present and compare the results of two different explainability techniques—Grad-CAM, SHAP and LIME applied in distinct domains. Each technique provides unique insights into the interpretability of AI models, and we assess their use based on the findings from two academic studies.

A. Study 1: Enhancing Breast Cancer Diagnosis in Mammography

The first paper [17] investigates the integration of Convolutional Neural Networks (CNNs) with XAI for improved breast cancer detection using the CBIS-DDSM dataset[1]. The study employs transfer learning with pre-trained networks such as VGG-16, Inception-V3, and ResNet50 while focusing on Grad-CAM, LIME, and SHAP to enhance interpretability.

The results of this study underscore the importance of interpretability in medical AI systems. The application of Grad-CAM, LIME, and SHAP on mammographic images reveals how the model discerns between benign and malignant findings.

a) Grad-CAM: Grad-CAM heatmaps (see Fig. 1) illustrate regions of diagnostic importance by highlighting areas with red hues, indicating heightened relevance. The heatmaps align with denser tissue or anomalies, reinforcing the model's ability to detect subtle but significant patterns. Figure 1 demonstrates how these heatmaps correspond with regions that require closer examination.

Strengths:

- Intuitive Visualization: The heatmaps generated by Grad-CAM visually highlight areas of diagnostic significance (with colors like red indicating the most critical zones), aiding in understanding for medical professionals.
- Alignment with Clinical Practices: The highlighted areas often correspond to dense tissue or anomalies relevant for diagnosis, providing confirmations to human experts.
- Enhances Trust in Models: The ability to clearly visualize where the model focuses its attention improves confidence in the results produced by Artificial Intelligence.

Weaknesses:

- Limited to CNNs: Grad-CAM is specific to convolutional neural networks and cannot be transferred to other architectures.
- Limited Resolution: The heatmap output can provide a coarse view, lacking detailed pixel-level explanations.

[1]https://www.kaggle.com/datasets/awsaf49/cbis-ddsm-breast-cancer-image-dataset

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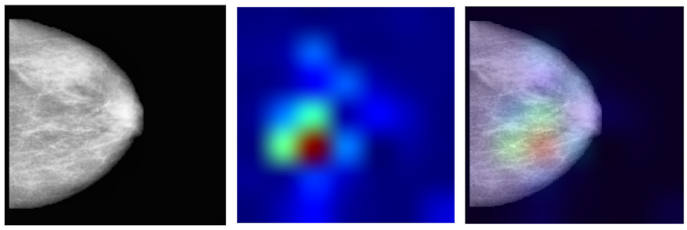


Fig. 1. Grad-CAM results (Left column: Original image, Center column: heatmap generated by Grad-CAM, Right column: heatmap superimposed on image)

b) LIME: The LIME technique provides further segmentation by dividing the mammogram into segments of varying relevance to the model's prediction (Fig. 2). It highlights not only tissue density but also unusual textures, confirming the relevance of features that align with expert observations.



Fig. 2. LIME results: Left—Original image; Center—Segmentation; Right—Important segments overlaid.

Strengths:

- Model-Agnostic: LIME can be applied to various machine learning models, not limited to CNNs.
- Explanatory Segmentation: It divides images into segments of varying relevance, allowing for a more granular understanding of which areas most influence classification decisions.
- Visual Validation: The areas identified by LIME often coincide with regions that a radiologist might find suspicious, confirming the model's reliability.

Weaknesses:

- Variability in Results: LIME may generate inconsistent explanations, varying the segmentation masks based on the samples used. This lack of stability can compromise clinical reliability.
- Computational Intensity: The analysis with LIME requires many iterations, making it less suitable for real-time applications.

c) SHAP: SHAP values offer a granular view of individual pixel contributions to the model's output (Fig. 3). Through a color-coded scheme, SHAP assigns red to features indicative of malignancy and blue to benign features, providing a direct link between AI decisions and clinically relevant markers.

Strengths:

- Detailed Interpretation: SHAP provides granular explanations of how each pixel (or feature) contributes to the final outcome, enabling in-depth analysis.

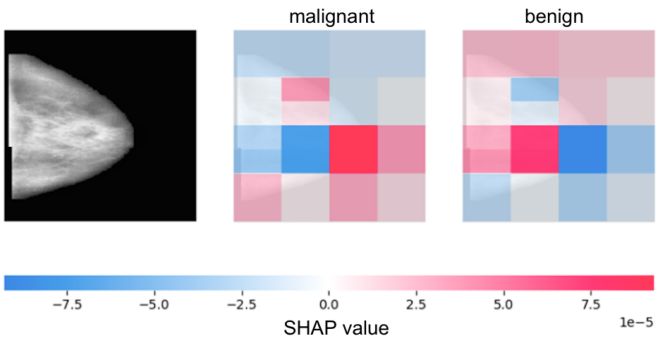


Fig. 3. SHAP results: Influence of individual features on the model's decision.

- Quantitative Alignment: SHAP values not only confirm the presence of malignancy but also correlate with clinically relevant markers, increasing expert trust.
- Mathematical Consistency: Being based on game theory, SHAP ensures a fair distribution of importance among features.

Weaknesses:

- Computationally Expensive: Calculating SHAP values is resource-intensive and time-consuming, making it challenging to use in real-time clinical scenarios.
- Interpretive Complexity: Although the values are detailed, they may be difficult for users without a solid background in XAI and machine learning to interpret.

This study demonstrates that combining CNNs with Grad-CAM, LIME, and SHAP enhances not only the diagnostic accuracy of AI models but also the interpretability of their decisions. Using XAI fosters trust between AI systems and medical professionals, promoting a more effective integration of AI technologies in clinical practice. However, limitations remain, particularly with the consistency of LIME, suggesting a need for ongoing refinement in XAI methodologies.

B. Study 2: SHAMSUL: Systematic Holistic Analysis to Investigate Medical Significance Utilizing Local interpretability methods in deep learning for chest radiography pathology prediction

The SHAMSUL study [18] utilized 234 input data points to evaluate model performance, with a weighted average AUROC score of 0.76 and a weighted average AUPRC score of 0.65. Notably, the Fracture class was excluded due to a lack of positive instances. The model demonstrated robustness even in the face of class imbalance, particularly excelling in detecting the Pleural Other class despite having only a single positive instance. This highlights the model's effectiveness in accurately identifying pathologies across various distributions.

a) Grad-CAM: Figure 4 highlights Grad-CAM's ability to cover relevant areas but also illustrates the challenge of broad segmentations that may obscure precise boundaries.

Strengths:

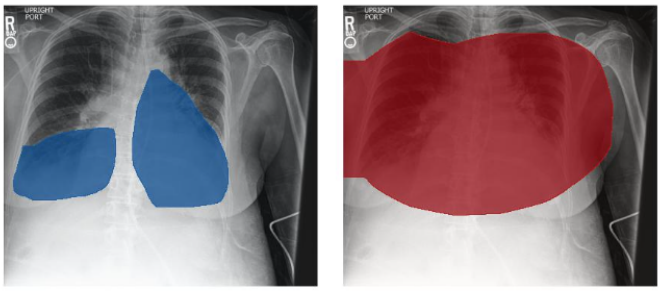
- Intuitive Visualization: Grad-CAM generates visual heatmaps highlighting key areas of the analyzed images,

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making it easier for clinicians to interpret model decisions Balanced.
- Performance: It exhibits a balanced performance across many pathology classes, achieving reasonable IoU scores for various conditions.

Weaknesses:

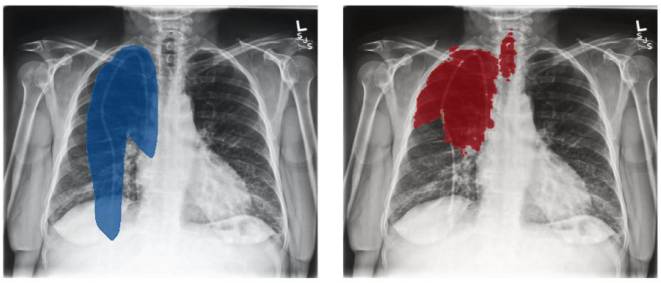
- Excessive Coverage: Grad-CAM's segmentation tends to be broad, which can obscure the precise boundaries of relevant regions, complicating clinical interpretation.
- Lack of Sptimes fails to identify relevant areas, as evidenced by a zero IoU score for the Lung Lesion class, indicating poor correspondence with expert annotations.



(a) Ground Truth (b) Grad-CAM Seg.

Fig. 4. Grad-CAM heatmap score segmentation of the Lung Opacity (IoU score: 0.35, prediction probability score: 0.54) class with human expert segmentation annotation for comparison.

b) LIME: Figure 5 illustrates how LIME effectively identified areas of interest but occasionally included irrelevant regions, demonstrating its potential and limitations in accurately localizing pathologies.



(a) Ground Truth (b) LIME Segment.

Fig. 5. LIME heatmap score segmentation of the best IoU scoring Support Devices (IoU score: 0.48, prediction probability score: 0.86) class instance with human expert segmentation annotation for comparison.

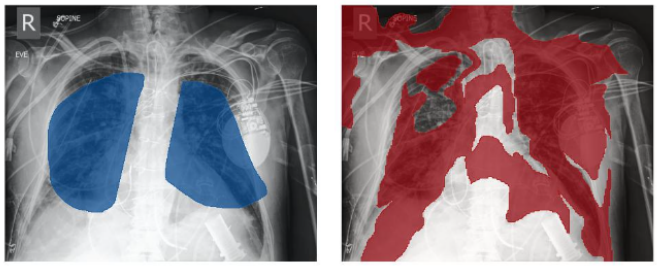
Strengths:

- Focused Segmentation: LIME can identify specific and pertinent areas in images, which is beneficial for clinicians in localizing pathologies.
- Promising Performance Metrics: It demonstrates favorable scores for specific classes, such as Edema, highlighting its potential effectiveness in targeted scenarios

Weaknesses:

- Inconsistency in Results: The variable scores suggest that LIME may struggle to generalize well across all pathology classes.
- Inclusion of Irrelevant Areas: LIME occasionally identifies funding artefacts, which can compromise its accuracy.

c) SHAP: Figures 6 indicate that while SHAP can cover relevant regions, its extensive coverage often leads to a lack of specificity, diminishing its interpretative value due to the erroneous inclusion of unrelated structures.



(a) Ground Truth (b) SHAP Segment.

Fig. 6. SHAP heatmap score segmentation of the Lung Opacity (IoU score: 0.34, prediction probability score: 0.52) class with human expert segmentation annotation for comparison.

Strengths:

- Detailed Contribution Analysis: SHAP provides detailed explanations regarding the contributions of each feature to the model's outcome, enhancing transparency.
- Accuracy in Relevant Areas: In certain cases, SHAP effectively identifies and localizes areas of interest.

Weaknesses:

- Limited Generalization: It has not shown robustness across all classes, with null IoU scores for several pathological categories.
- Excessive Coverage: The SHAP segmentation tends to cover large areas, leading to a loss of specificity and discriminative power.

Grad-CAM demonstrates its utility by providing intuitive visualizations that enhance the understanding of model predictions, particularly for conditions like Cardiomegaly. However, its broad coverage often obscures finer boundaries, which can be critical in clinical settings.

LIME shows promise in identifying relevant regions in images, yet its performance can be inconsistent, leading to the inclusion of irrelevant areas. This inconsistency emphasizes the need for refinement to improve its reliability in medical applications. Lastly, SHAP, while capable of delivering detailed insights, often suffers from excessive coverage that can dilute interpretive clarity.

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V. Discussion

Integrating XAI methods like Grad-CAM, LIME, and SHAP into medical imaging presents distinct advantages and challenges. Each technique serves unique purposes in interpreting model outputs, and their effectiveness can vary based on the specific application and clinical context.

Grad-CAM offers immediate visualizations that are intuitive and beneficial for clinicians. Its strength lies in generating heatmaps that highlight areas of interest in high-resolution medical images, such as MRIs and X-rays. However, Grad-CAM's limitations include its reliance on Convolutional Neural Networks (CNNs), which restricts its applicability to other model types. Furthermore, it may lack fine-grained precision, leading to ambiguous interpretations in complex cases. In contrast, model-agnostic techniques like LIME provide greater versatility across different models and medical image types. Its model-agnostic nature allows it to be applied broadly; however, the need for perturbation sampling can be time-consuming and lead to variability in results. This variability necessitates careful consideration and potentially cautious application in clinical settings, where consistent and reliable explanations are crucial.

SHAP delivers detailed and mathematically consistent insights into model predictions. Its ability to provide both local and global interpretability across various models adds to its appeal. Nonetheless, SHAP is computationally intensive and may pose challenges for real-time applications due to its complexity. This makes it less suitable for environments where time-sensitive decisions are critical.

VI. Opportunities and Future Directions

Several important directions should be explored to advance the development of explainable artificial intelligence (XAI) in medical imaging techniques. A key area for improvement lies in visual explanations, especially in techniques like Grad-CAM. By enhancing the outputs' resolution and clinical relevance, XAI methods could become more actionable, allowing clinicians to interpret the results and integrate them into patient care easily. Improved visual feedback would bridge the gap between complex machine learning models and the practical needs of healthcare professionals. Another promising avenue is real-time integration. Advances in computational efficiency may soon enable complex methods like SHAP to operate in real-time, making them more suitable for clinical workflows. Achieving this would foster greater trust in AI systems and facilitate clinicians' adoption, as time-sensitive environments demand swift and reliable decision-making tools. A valuable strategy for the future involves hybrid approaches—combining different XAI techniques to gain quick insights and deeper understanding. For instance, Grad-CAM could offer a fast, intuitive overview of the model's predictions, while SHAP could provide detailed, granular explanations. This complementary use of methods would allow healthcare professionals to balance efficiency with precision, tailoring the interpretability tools to different clinical scenarios. Lastly, it is crucial to collaborate closely with clinicians throughout the development process. Future research must prioritize practitioner feedback to refine these techniques, ensuring that XAI tools meet the practical demands of healthcare settings. This collaboration will help align technological advancements with the realities of clinical workflows, ultimately increasing the effectiveness and usability of AI-based tools in medical environments.

VII. Conclusion

In conclusion, while Grad-CAM, LIME, and SHAP each provide valuable insights, they exhibit distinct strengths and limitations that affect their suitability in clinical environments. Grad-CAM is appreciated for its intuitive visual heatmaps but is restricted to CNNs and can lack precision in complex cases. LIME offers flexibility across diverse models but may produce inconsistent results due to the stochastic nature of its sampling. SHAP delivers detailed and accurate explanations, but its computational intensity can hinder real-time application. Advancing these methods to achieve higher specificity, clarity, and efficiency will be critical for their seamless integration into clinical workflows, enhancing patient care and outcomes. Ultimately, fostering trust and collaboration between AI systems and medical professionals will hinge on the development of more accessible and precise interpretability techniques.

References

[1] Selvaraju, R. R., Cogswell, M., Das, A., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. Proceedings of the IEEE International Conference on Computer Vision (ICCV).

[2] Guidotti, Riccardo. "Counterfactual explanations and how to find them: literature review and benchmarking." Data Mining and Knowledge Discovery 38.5 (2024): 2770-2824.

[3] Yoo, Soyoung, and Namwoo Kang. "Explainable artificial intelligence for manufacturing cost estimation and machining feature visualization." Expert Systems with Applications 183 (2021): 115430.

[4] J. A. Recio-Garcia, B. Diaz-Agudo, and V. Pino-Castilla, "CBR-LIME: a case-based reasoning approach to provide specific local interpretable model-agnostic explanations," in Case-Based Reasoning Research and Development: 28th International Conference, ICCBR 2020, Salamanca, Spain, June 8–12, 2020, Proceedings 28, pp. 179–194, Springer, 2020.

[5] E. M. Kenny, C. Ford, M. Quinn, and M. T. Keane, "Explaining black-box classifiers using post-hoc explanations-by-example: The effect of explanations and error-rates in XAI user studies," Artificial Intelligence, vol. 294, p. 103459, 2021.

[6] F. Emmert-Streib, O. Yli-Harja, and M. Dehmer, "Explainable artificial intelligence and machine learning: A reality rooted perspective," Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, vol. 10, no. 6, p. e1368, 2020.

[7] H. Chereda, A. Bleckmann, K. Menck, J. Perera-Bel, P. Stegmaier, F. Auer, F. Kramer, A. Leha, and T. Beißbarth, "Explaining decisions of graph convolutional neural networks: patient-specific molecular subnetworks responsible for metastasis prediction in breast cancer," Genome medicine, vol. 13, pp. 1–16, 2021.

[8] A. B. Arrieta, N. Diaz-Rodriguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. Garcia, S. Gil Lopez, D. Molina, R. Benjamins, et al., "Explainable artificial intelligence XAI: Concepts, taxonomies, opportunities and challenges toward responsible ai," Information fusion, vol. 58, pp. 82–115, 2020.

[9] van der Waa, Jasper, et al. "Evaluating XAI: A comparison of rule-based and example-based explanations." Artificial intelligence 291 (2021): 103404.

[10] Ntrougkas, Mariano, Nikolaos Gkalelis, and Vasileios Mezaris. "Tame: Attention mechanism based feature fusion for generating explanation maps of convolutional neural networks." 2022 IEEE International Symposium on Multimedia (ISM). IEEE, 2022.

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[11] Staffa, Mariacarla, and Paola Barra. "Multi-monitor System for Adaptive Image Saliency Detection Based on Attentive Mechanisms." International Conference on Human-Computer Interaction. Cham: Springer Nature Switzerland, 2023.
[12] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in Proceedings of the IEEE international conference on computer vision, pp. 618–626, 2017.
[13] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. Advances in neural information processing systems, 30, 2017.
[14] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should I trust you?" Explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, pages 1135–1144, 2016.
[15] Qi, Zhongang, Saeed Khorram, and Fuxin Li. "Visualizing Deep Networks by Optimizing with Integrated Gradients." CVPR workshops. Vol. 2. 2019.
[16] Das, Arun, and Paul Rad. "Opportunities and challenges in explainable artificial intelligence (xai): A survey." arXiv preprint arXiv:2006.11371 (2020).
[17] Ahmed, Maryam, et al. "Enhancing Breast Cancer Diagnosis in Mammography: Evaluation and Integration of Convolutional Neural Networks and Explainable AI." arXiv preprint arXiv:2404.03892 (2024).
[18] Alam, Mahbub Ul, et al. "SHAMSUL: Systematic Holistic Analysis to investigate Medical Significance Utilizing Local interpretability methods in deep learning for chest radiography pathology prediction." Nordic Machine Intelligence 3 (2023): 27-47.
[19] Amin, Al, et al. "An explainable ai framework for artificial intelligence of medical things." 2023 IEEE Globecom Workshops (GC Wkshps). IEEE, 2023.

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